

Mojtaba Hajian

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Profile

Ph.D. in Information and Communication Technology (ICT) specializing in **Big Data Analytics and Distributed Data Systems**, with a research background in **scalable analytics, large-scale OLAP systems, and multidimensional data cube compression**. Extensive experience in designing and optimizing **high-volume data processing pipelines** using **Apache Spark, Hive, and Hadoop**, with a focus on efficient query processing, dimensionality reduction, and performance benchmarking over massive datasets. Proficient in applying **machine learning and unsupervised analytics on top of big data platforms** to extract actionable knowledge from complex, high-dimensional data. Experienced in cloud-based data architectures, data warehousing concepts, and research-grade big data solutions, supported by a publication record in Big Data and OLAP systems.

Experience

- **PH.D. CANDIDATE | UNIVERSITY OF CALABRIA, DIMES, ITALY | JANUARY 2022 – MARCH 2025**
 - Develop large-scale data analytics workflows utilizing Apache Spark, Hive, and Hadoop to support high-volume, low-latency data processing
 - Integrating ML on top of distributed data platforms
 - Leverage unsupervised learning methods (e.g., K-Means, DBSCAN) to discover patterns within complex, high-dimensional data
 - Apply OLAP techniques for effective knowledge extraction from compressed data cubes in big data ecosystems
 - Author scientific publications, technical reports, and conference papers, effectively communicating research insights to multidisciplinary audiences
 - Utilize Python, Matlab, C#, and Java for data engineering, modeling, and analytical tasks
- **RESEARCH INTERN | ECOLE D'INGÉNIEURS GÉNÉRALISTES, DE L'INFORMATIQUE À L'ÉLECTRONIQUE (ESEO), FRANCE | JUNE 2023 – DECEMBER 2023**
 - Implemented big data frameworks: Successfully deployed Apache Hadoop, Apache Hive, and Apache Spark to establish a big data processing environment
 - Database Management: Created and managed a TPC-H database for benchmarking OLAP processes, integrating Apache Hive for structured data storage and manipulation
 - OLAP Data Cube Deployment: Deployed and managed an OLAP data cube in Hive, enabling advanced data analysis capabilities
 - Machine Learning Algorithms Deployment: Applied machine learning algorithms over big OLAP data cubes
- **RESEARCH FELLOW | FREE UNIVERSITY OF BOZEN-BOLZANO, FACULTY OF ENGINEERING, ITALY | JUNE 2025 – PRESENT**
 - Working on the AIRFRAME project on AI-driven radiomics for metastatic castration-resistant prostate cancer (mCRPC), analysing *multi-modal PET/CT images* to predict therapy outcome and survival

- Design and validate *machine-learning and deep-learning models*
 - Radiomics features extraction pipelines from medical images and high-dimensional data reduction
 - Implement explainable-AI techniques to visualize feature importances and patient-specific outcome probabilities for clinicians
 - Writing professional reports and publishing in peer-review journals
- **SIGNAL PROCESSING AND RECORDING SPECIALIST | NATIONAL BRAIN MAPPING LAB (NBML), IRAN | APRIL 2019 – JANUARY 2022**
 - Led a team as Head of EEG and Signal Processing Department; designed experimental protocols for multi-modal EEG acquisition and analysis
 - Developed protocols for multi-modal EEG data acquisition and signal processing
 - Designed experimental tasks using Psychtoolbox and PsychoPy
 - Analyzed EEG data using EEGLAB, MATLAB Toolboxes, and Python MNE

Education

- **PH.D. IN INFORMATION AND COMMUNICATION TECHNOLOGY (ICT) | MARCH 2025 | UNIVERSITY OF CALABRIA, ITALY**
Dissertation: *Multidimensional Big Data Analytics for Knowledge Discovering and Analysis from Big Web Knowledge Bases*
 - Focus on scalable analytics, OLAP systems, data cube compression**Supervisor:** Prof. Alfredo Cuzzocrea
- **M.S. IN BIOMEDICAL ENGINEERING | JULY 2017 | AMIRKABIR UNIVERSITY OF TECHNOLOGY, IRAN**
Dissertation: *Quantification of Depression Disorder Using EEG Signal*
Supervisor: Prof. Mohammad Hassan Moradi
- **B.S. IN ELECTRONICS ENGINEERING | SEPTEMBER 2014 | BABOL NOSHIRVANI UNIVERSITY OF TECHNOLOGY, IRAN**
Dissertation: *Design and Implementation of A Digital Compass*
Supervisor: Prof. Mehdi Ezoji

Skills

- **Big Data & Data Platforms:** Apache Spark, Hive, Hadoop, OLAP systems, distributed data processing, data warehousing concepts, scalable ML
- **Data Analytics & Machine Learning:** Supervised/Unsupervised Learning, Artificial Neural Networks (Scikit-learn), classical ML (NNs, SVM, xGBoost, Random Forest), Explainable AI (SHAP, feature importance)
- **Deep Learning:** CNN, RNN, LSTM, GNN, Reinforcement Learning, Transformers (TensorFlow, Keras, Pytorch), LLMs (LLaMA, LangChain, RAG)
- **Programming:** Python, Matlab, Java, C#
- **Dev/Research Workflow:** Git, Linux, Docker

Publications

1. [Submitted] Alfredo Cuzzocrea, Mojtaba Hajian, “**Multidimensional Clustering over Big Data: Models, Issues, Analysis, Emerging Trends**”, *Journal of Big Data*, Springer, 2026.

2. [In Progress] Alfredo Cuzzocrea, Mojtaba Hajian, “**Data Cube Compression at Work! Effective and Efficient OLAP over Big Datasets in Mobile Cloud Environments**”, *Information Systems*, 2026.
3. Alfredo Cuzzocrea, Mojtaba Hajian, “**ClassCube: Effective and Efficient Big OLAP Data Cube Classification via Dimensionality Reduction**”, *ICEIS*, 2025.
4. Alfredo Cuzzocrea, Mojtaba Hajian, Abderraouf Hafsaoui, “**MALAGA – MultidimensionAL Big Data Analytics over Massive Graph Data**”, *IEEE International Conference on Big Data* (2024), Washington DC, USA, 2024.
5. Alfredo Cuzzocrea, Mojtaba Hajian, “**Experimental Analysis and Assessment of a Real-Life Cloud Mobile Big OLAP System Enhanced with Compression and Approximation Paradigms**”, *IEEE International Conference on Big Data* (2024), Washington DC, USA, 2024.
6. Alfredo Cuzzocrea, Mojtaba Hajian, Abderraouf Hafsaoui, “**Compressing Big OLAP Data Cubes over Mobile Clouds: A Hierarchy-Based Data Partitioning Approach**”, *Proceedings of the 32nd Symposium of Advanced Database Systems, Villasimius, Italy*, 2024.
7. A Cuzzocrea, P Figuera, M Hajian, PG Bringas, ‘**A Theoretical Framework for Supporting Clustering Validation via Non-Negative-Matrix-Factorization Trace Sequences Over Probabilistic Spaces**’, *IEEE Conference on Cloud and Big Data Computing (CBDDCom 2023)*, Abu Dhabi, United Arab Emirates.
8. Alfredo Cuzzocrea, Carson Leung, Mojtaba Hajian and Marshall Jackson, ‘**Effectively and Efficiently Supporting Predictive Big Data Analytics over Open Big Data in the Transportation Sector: A Bayesian Network Framework**’, *The 8th IEEE International Conference on Cloud and Big Data Computing (CBDDCom 2022)*, Calabria, Italy.
9. Alfredo Cuzzocrea, Christophe Cerisara, Mojtaba Hajian, ‘**Risk Analysis for Unsupervised Privacy-Preserving Tools**’, *Proceedings of the 30th Italian Symposium on Advanced Database Systems, SEBD 2022, Tirrenia (Pisa)*, Italy, June 19-22, 2022.
10. Mojtaba Hajian, Mohammad Hassan Moradi, ‘**Quantification of Depression Disorder Using EEG Signal**’, *The 24th National and the 2nd International Iranian Conference on Biomedical Engineering (ICBME 2017)*, Tehran, Iran, 2017.
11. Yousef Mohammadi, Mojtaba Hajian, Mohammad Hassan Moradi, ‘**Discrimination of Depression Levels Using Machine Learning Methods on EEG Signals**’, *The 27th Iranian Conference on Electrical Engineering (ICEE 2019)*, Yazd, Iran, 2019.
12. [In Progress] Mojtaba Hajian, Somayeh Khosravi Khorashad, Sepideh Khoneiveh, Reza Khosrowabadi, “**Decoding Neural Dynamics of Suspense: EEG Connectivity and Power Analysis with Machine Learning for Pre- and Post-Suspense State Classification**”, *Applied Soft Computing*, 2026.
13. Somayeh Khosravi Khorashad, Sepideh Khoneiveh, Mojtaba Hajian, Reza Khosrowabadi, ‘**Decoupage Parameters Related to the Suspense Points and Symmetrical Activation of Brain Waves at the Frontal Lobe: An EEG Study of the Movie ‘Psycho’**’, *Quarterly Review of Film and Video*, 2022.
14. A Cuzzocrea, M Hajian, ‘**Effective and Efficient Big OLAP Data Cube Compression in Mobile Cloud Environments: The IQTS Algorithm**’, *IEEE International Conference on Big Data* (2023), Sorrento, Italy.

Teaching Experiences

- “Knowledge Transfer Activity at Paracelsus Medical University (Salzburg)”, Workshop, Provided methodological training and support to medical physics experts, Salzburg, Austria, September 2025.
Topics Covered:
 - Image preparation and automated lesion segmentation workflow
 - Radiomics feature extraction
 - Feature organization and dataset construction
 - Machine learning modeling strategies for small datasets
- “Machine Learning Applications for Biomedical Signal Processing”, Webinar, Brain Mapping Laboratory, Tehran, Iran, February 2021.
- “The 16th workshop on EEG Signal Recording, Processing and Analyzing”, Workshop, National Brain Mapping Laboratory, Tehran, Iran, October 2020.
- “Instructor of over 50 trainees in the EEG data analysis and recording internship course”, Internship, National Brain Mapping Laboratory, Tehran, Iran, August 2019-December 2021.
- “An introduction to the function of the EEG amplifier and how to use it”, Seminar, Brain Mapping Laboratory, Tehran, Iran, January 2020.
- “Robotics and micro-controllers” Workshop, Babol Noshirvani University of Technology, Mazandaran, Iran, July 2010-September 2013.

Honors and Awards

- Commendation for Establishing EEG Setup for Sleep Services
 - National Brain Mapping Laboratory (NBML) | 2020
- Team Leader Award for Manufacturing Electronic MRI-Compatible Response Grips
 - National Brain Mapping Laboratory | 2020
- Reviewer for 'Frontiers in Biomedical Technologies' Journal and 'IEEE Big Data' Conference
- 2nd Place in Smart City Startup Weekend (MOPEC-2017)
 - Sponsored by Municipality of 5th District of Tehran City | 2017

References

- Dr. Floriano Zini
 - Senior Researcher of Computer Science
 - Free University of Bozen-Bolzano
 - Email: floriano.zini@unibz.it
- Prof. Alfredo Cuzzocrea
 - Professor of Computer Engineering
 - University of Calabria
 - Email: alfredo.cuzzocrea@unical.it
- Prof. Slimane Hammoudi
 - Professor at Computer Science Department
 - ESEO, France
 - Email: Slimane.HAMMOUDI@eseo.fr
- Dr. Reza Khosrowabadi
 - Professor at the Institute for Cognitive and Brain Sciences
 - Shahid Beheshti University

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